

Tribo Tribe: Triboelectric Interaction Sensing with 3D Physical Interfaces

Xin Liu

liuxin@u.nus.edu

Division of Industrial Design/Keio-NUS CUTE Center
National University of Singapore
Singapore

Clement Zheng

clement.zheng@nus.edu.sg

Division of Industrial Design/Keio-NUS CUTE Center
National University of Singapore
Singapore

Bo Han

e0787995@u.nus.edu

Division of Industrial Design
National University of Singapore
Singapore

Ching Chiuan Yen

didyc@nus.edu.sg

Division of Industrial Design/Keio-NUS CUTE Center
National University of Singapore
Singapore



Figure 1: *Tribo Tribe* Applications: A) Six interactive widgets (press, pull, toggle, slide, flip, rotate) and an integrated game controller. B) ConductTENG: a contact-less music device. C) Cattery: a mood lighting system. D) Dot a Spot: an interactive educational tool.

ABSTRACT

This research introduces “*Tribo Tribe*”, a technique for fabricating 3D tangible interactive interfaces capable of sensing movement inputs through ubiquitous materials. *Tribo Tribe* is built on the working principle of Triboelectric Nanogenerators (TENG) to enable self-powered sensing to 3D systems. We introduce a tool kit that facilitates designers and makers to easily customise both prototyping and sensing through TENG technology. We also demonstrate four design possibilities for different fields to illustrate how *Tribo Tribe* can instrument TENG into 3D physical interactive prototypes (Figure 1).

CCS CONCEPTS

• Human-centered computing → Interactive systems and tools.



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KEYWORDS

Tangible Interaction, 3D Systems, TENG, Self-powered Sensing, ubiquitous materials.

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1 INTRODUCTION

Many of us live in intelligent and interactive environments enabled by ubiquitous computing [22]. Within this paradigm, there is a significant body of research surrounding TUIs (Tangible User Interfaces) [8]. TUIs seek to push HCI beyond graphical user interfaces and conventional input devices; leveraging the richness of actions and materials in the physical world for computational interaction and interfaces. Over the last two decades, the advance and democratization of personal fabrication [2] and physical computing technology lowered the barrier for designers to imagine and realize functional TUIs. However, prototyping TUIs remains a challenging electro-mechanical task [11] for many designers. Different interactions require different electronic parts to work. These prefabricated components come in different sizes and forms. Connecting

and embedding these parts into a bespoke physical interface is a complicated puzzle that designers grapple with.

With this challenge in mind, we looked into Triboelectric Nanogenerators (TENG). TENG is a technology that converts physical movements to electricity through the triboelectric effect. The triboelectric effect (typically known as “static electricity”) is ubiquitous in our everyday life and results from two materials coming into contact and separating with each other. It is generally regarded as a problematic phenomenon in the manufacturing industry given that electrostatic charges induced can lead to ignition, dust explosions, dielectric breakdown, and electronic damage [25]. However, from the perspective of TUIs, the electrical signals translated from mechanical movements through the triboelectric effect can be used for energy harvesting [9] or activity sensing [1, 5].

We are motivated to facilitate designers to easily embed sensing into 3D interactive systems. In this paper, we propose *Tribo Tribe*. *Tribo Tribe* builds on the principle of TENG, turning static electricity into triboelectric signals through commonly found materials. We introduce four material pairs that are easily applied to 3D systems as both triboelectric and physical prototyping materials. We also showcase design examples enabled by *Tribo Tribe*. These examples demonstrate tangible user interfaces such as a contact-less music device, a mood lighting system, and an interactive educational tool. Compared to embedding external electronics, *Tribo Tribe* makes use of the existing mechanical movements in these interfaces to detect physical interactions without off-the-shelf sensors.

2 RELATED WORK

This work builds upon research on prototyping tangible interfaces with input sensing, as well as applying TENG for tangible interaction.

2.1 Prototyping Tangible Interfaces with Input Sensing

One approach to making prototypes interactive is using a single approach to instrument an interface to sense different tangible interactions, such as Printed Optics [24], Sauron [11], 3D Printing Pneumatic Device Controls [18], and Machamagnets [28]. These approaches simplify the inventory required for building functional devices. However, they still require designers and makers to design complicated assemblies comprising 3D printed parts, electronics, and other supporting components (e.g., magnets).

Other researchers put the sensors outside the 3D printed models to avoid this problem. For example, a typical fabrication approach is removing interior material from 3D models to form tubes, pipes, or cavities, e.g., 3D printing pneumatic device [18], AirTouch [17], Blowhole [16], A series of tubes [12]. Although such approaches do not require embedded sensors, they still require external sensors that take up space in the overall system. To address problems of space and electrical connections, researchers directly fabricate circuits via conductive filaments through multi-material 3D printing (PrintPut [4], Capricate [13], /trilaterate [15], MetaSense [7], and Flexibles [14]).

Inspired by these projects, we explored using off-the-shelf materials to directly instrument 3D printed parts for sensing tangible

interaction. *Tribo Tribe* explores TENG technology to directly convert the mechanical energy generated during user interaction with the tangible input into electrical signals.

2.2 TENG for Tangible Interaction

HCI researchers have leveraged TENG technology for tangible interaction through readily available materials. Paper generators [9] proposed interactive energy harvesters that generated electrical energy from the user interactions with paper-like materials, such as polytetrafluoroethylene (PTFE) films and aluminium foil. SATURN [1] used PTFE films and copper tape for self-powered microphone sensors, while SPIN [5] used the same materials for sensing origami structures. These work focused on applying TENG to interfaces fabricated from paper-like materials. We see an opportunity to extend TENG for HCI applications to 3D interfaces. We draw inspiration from other fields that demonstrate applying TENG to 3D systems for energy harvesting [10] or sensing for specific applications [6, 20].

Tribo Tribe extends TENG to 3D interactive systems broadening the physical expression of such interfaces within HCI.

3 TRIBO TRIBE: DESIGN EXPLORATION

3.1 Sensing Principle

Fundamentally, *Tribo Tribe* builds on the single-electrode TENG configuration [21] shown in Figure 2. The electrode (grey) is connected to ground, while the interactive (moving) part forms the dielectric layer (blue). This configuration offers more flexibility for tangible interactions as the interactive region of the sensing system remains untethered compared to the double-electrode modes [19]. The quality of the triboelectric signal depends on two key factors: 1) the triboelectric potential of a material pair (specifically, a conductive material and a polymer material), and 2) the relative movement between the two materials.

Essentially, when the dielectric layer (blue) and electrode (grey) approach each other and come into contact (Figure 2 A, B), a chemical bond is formed. During this adhesion, charges move from electrode to ground to equalise their electrochemical potential [21]. This results in an instantaneous positive voltage. Upon separation (Figure 2 C, D), the electrode material tends to hold extra electrons from the ground to maintain charge balance, resulting in an instantaneous negative voltage. Therefore, triboelectric signals have an AC signal pattern that consistently exhibits both positive and negative values; for example, the AC signal generated by nickel conductive coating and acrylic polymer coating is shown in Figure 3 B.

3.2 Toolkit

We developed a toolkit (Figure 3 A) for designers to easily apply TENG technology to 3D systems. This toolkit consists of triboelectric materials (i.e., four pairs of conductive and polymer materials), customised PCBs for signal processing, and open-source physical computing platforms (we demonstrate this toolkit with the Arduino Nano Every microcontroller in this paper).

3.2.1 Conductive and Polymer Materials for TENG sensing. Researchers have evaluated many materials for their triboelectric effect, notably a study of 50 polymer film materials by [29]. In *Tribo*

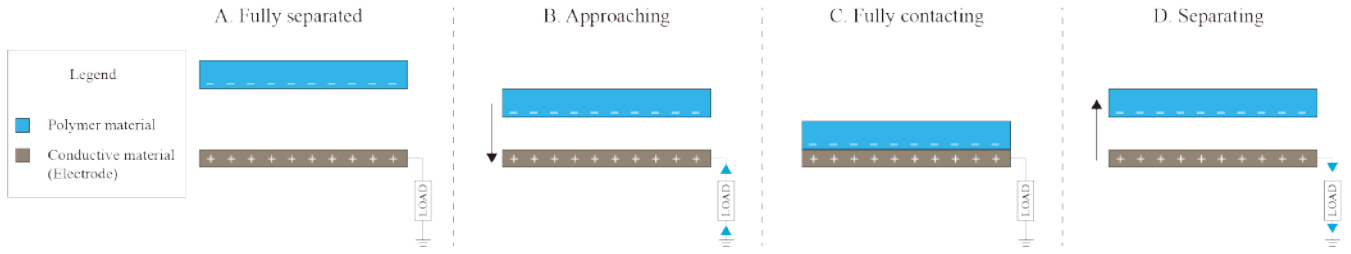


Figure 2: The Working Principle of Single-electrode Mode.

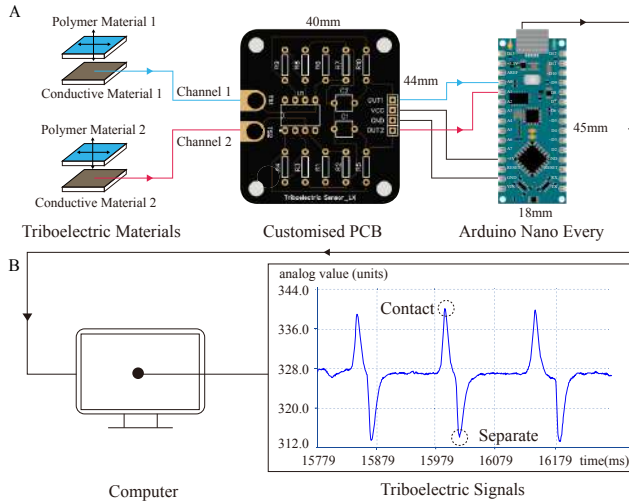


Figure 3: Toolkit and signal: A) Electronics connection diagram. B) AC signal generated from nickel conductive spray paint (841R) and acrylic conformal spray paint from MG Chemicals.

Tribe, we are interested in instrumenting 3D tangible interfaces with TENG sensing. We investigated materials that were more conveniently applied to 3D forms (in contrast to the well-studied 2D/film-based materials for TENG).

Inspired by Living Wall [3], Sprayable User Interfaces [23], Electrick [26], and Wall++ [27], we explored spray painting in tandem with common prototyping materials as a means of applying functional layers onto 3D surfaces. In detail, we experimented with four material-pairs for TENG sensing—a conductive material (nickel conductive spray paint) paired with four different polymer materials (acrylic spray paint, PTFE film, silicone, and polylactide (PLA) filament) (Table 1).

3.2.2 Customised PCB and Arduino Nano Every microcontroller for signal processing. We developed a PCB (Figure 4) to amplify and reduce the noise of the raw signal generated by the TENG sensors. This specific PCB offers two independent signal channels. Since the raw peak values generated were approximately 0.01 V , we used an operational amplifier (LM358P) with a gain of around 1000. We used resistor-capacitor circuit as low pass filters (upper limit: 22.5 Hz) to reduce noise. We used an Arduino Nano Every microcontroller to read the processed signal (Figure 3 B).

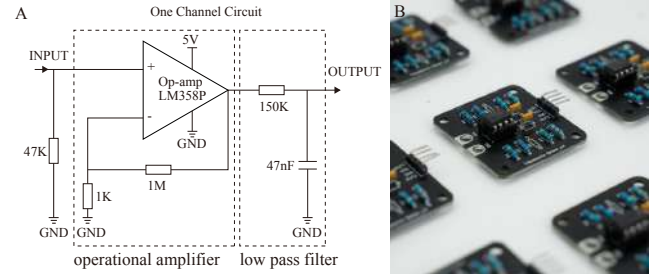


Figure 4: PCB design: A) One channel schematic diagram. B) Physical PCBs.

3.3 Design Space

With the *Tribo Tribe* toolkit, we explored the design space of TENG-enabled 3D interactive interfaces. We built a range of applications, including tangible user interfaces, contact-less gesture sensing, home products, and educational toys.

3.3.1 Input Widgets for Tangible User Interfaces. We developed six basic interactive widgets with a variety of input gestures, including press, pull, toggle, slide, flip, and rotate (Figure 5). These widgets are capable of detecting on/off states, keeping track of time, measuring interaction duration, or calculating rotation. These functional inputs are easily incorporated into a physical interface, such as a game controller (Figure 1 A). We offer these six widgets as 3D input primitives instrumented with TENG for others to customise.

These widgets were instrumented with nickel conductive spray paint and acrylic conformal spray paint as triboelectric materials. These widgets have a relatively small footprint of about 22 mm by 22 mm since no off-the-shelf sensors were embedded inside. The acrylic conformal coating covered the base of the moving parts (blue), and the nickel conductive spray was applied to the base of the enclosure (grey). User inputs cause these two coatings to contact and separate. Information such as time, count, and direction is computed from the raw signal by a microcontroller.

3.3.2 ConduTENG: Contact-less Gesture Sensing. ConducTENG is a sound control device inspired by music conductors who wave the baton to control the tempo of the entire band (Figure 6). Users choose a song and wave the “baton” across the stand to change the song’s tempo based on the frequency of the baton’s swing. This system challenges users to find the correct tempo of the selected song. Indicator lights on top of the stand provide feedback if the baton swing is too slow, too fast, or just right (Figure 6 C).

Table 1: Conductive and Polymer Material Descriptions.

Name	Brand	Color	Cure Time
Conductive Material			
Nickel Conductive Spray Paint (841AR)	MG Chemicals	Grey	24h @ 22°C; 30 min @ 65°C
Polymer Materials			
Acrylic Conformal Spray Paint	MG Chemicals	Transparent	24h @ 22°C; 30 min @ 65°C
Polytetrafluoroethylene (PTFE) Rolls	AIDMER	Translucent	N/A
Silicone	Dragon Skin 20	Translucent	4h @ 22°C
Polylactide (PLA) filament	Crealty	White	N/A

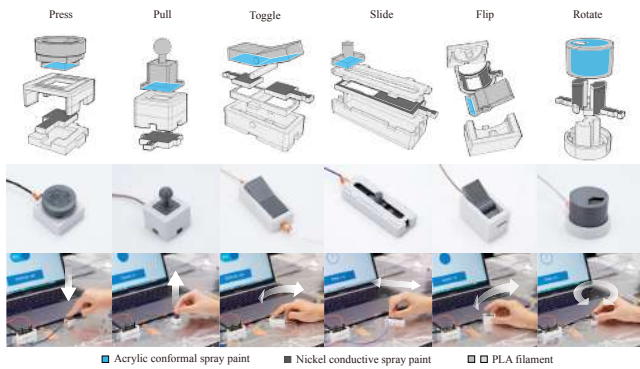


Figure 5: Input Widgets: Six interactive widgets with nickel conductive spray coatings and acrylic conformal spray coatings.

ConducTENG demonstrates a contact-less approach to detect physical gestures (i.e., the baton swing). The baton is constructed from PTFE rolls and generates triboelectric signals when moved relative to the stand—which contains an nickel conductive spray painted electrode. The materials used and the baton-stand embodiment enabled triboelectric signals to be generated despite the lack of physical contact.

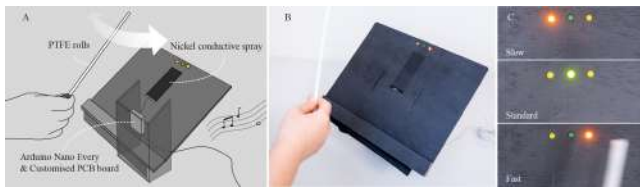


Figure 6: ConducTENG: A) The device structure and working materials (PTFE films and nickel conductive spray). B) The user controls music tempo by swinging the baton. C) Indicator lights: red light (too slow or fast) and green light (right).

3.3.3 Cattery: Mood Lighting Product. Cattery is a mood lighting product with a soft silicone outer shell (Figure 7). The cat's back, nose, and palms are all interaction points. Users touch the cat's back to increase the lamp's brightness, touch the nose to dim the light, and squeeze the hand to change colours. The silicone body served as a light diffuser as well as offered soft and elastic tangible interactions with the device.

More importantly, the silicone material itself served as the polymer material to trigger the TENG sensors. Nickel conductive spray

paint was applied onto the 3D curved surfaces inside the lamp. The silicone outer layer contacts and separates from the electrodes during interaction to generate a triboelectric signal. This entire device directly converts the mechanical energy of user interactions into triboelectric signals, enabling functional interaction without using commercial electronic sensors.

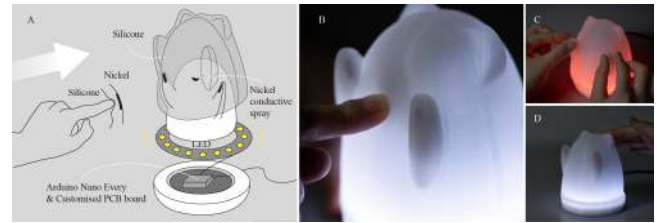


Figure 7: Cattery: A) Exploded view of the device. B) Brightening the light. C) Changing colours. D) Dimming the light.

3.3.4 Dot a Spot: Interactive Educational Tool. Dot a Spot is an educational system for children (Figure 8). It adds interactivity and entertainment to ordinary dot markers that are used to complete blank spots to form patterns like shapes and numbers. The Dot a Spot system includes the physical dot markers and coloring book, along with an accompanying digital book. Like existing dotting activities, children dot the spots at the corresponding positions of the physical book. The smart pen shell then recognises each stamp and triggers the related patterns on the digital book displayed on a screen. Dot a Spot aims to support children to experiment with mark making and artistic expression while practising hand-eye coordination and fine-motor control. The interactive system motivates children by adding more dynamic feedback and animation.

The smart pen is self-powered. It is 3D printed in PLA and also serves as the triboelectric polymer material. Here we sprayed nickel conductive paint as the triboelectric electrode on the inner surface of the pen shell. A triboelectric signal is generated each time children dot a spot.

4 BENEFITS & LIMITATIONS

Tribo Tribe extended tangible interactions with TENG and explored how it can be applied to 3D systems. At this stage, the contribution of *Tribo Tribe* can be summarised in the following ways:

- (1) *Tribo Tribe* broadens the design possibilities of TENG technology to 3D interactive systems (extending previous 2D

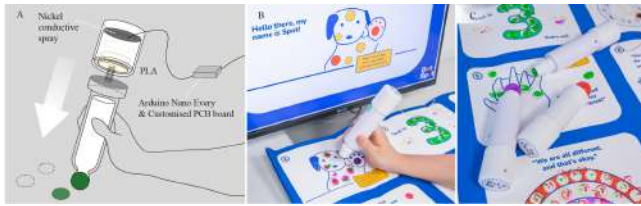


Figure 8: Dot a Spot: A) The smart pen's structure. B) Interacting with digital animations. C) Pens with different colours.

paper-based interfaces). Alongside sheet materials to achieve tangible interaction with TENG, we demonstrate different materials and fabrication approaches to efficiently instrument 3D physical interfaces.

- (2) We developed a simple *Tribo Tribe* toolkit. This toolkit comprises available commodity materials as well as custom electronics to harness TENG signals.
- (3) The application examples we demonstrate with *Tribo Tribe* addressed both prototyping and sensing in the design and fabrication of tangible interactive systems. We leveraged polymer materials in multiple ways for each application—as a prototyping *as well as* sensing material—as well as made use of conductive spray paints as an efficient way to apply electrodes onto 3D physical forms. We envision that this will lead to economical and electronics-saving tangible interfaces.

However, it is important for us to note the limitations of *Tribo Tribe* and TENG technology at large. *Tribo Tribe* is sensitive to environmental conditions. Triboelectric signals are weaker in environments with higher humidity, and it is also susceptible to electromagnetic interference from nearby devices. In certain scenarios, we also had to charge the TENG sensors, such as by moving the TENG materials back and forth a few times, to boost the signal. TENG signal strength is also easily affected by the wear and tear of materials.

5 FUTURE WORK

We explored four pairs of materials for triboelectric sensing and developed a range of interactive applications to demonstrate 3D tangible interactive systems enabled by TENG. Moving forward, we plan to explore more materials and conduct experiments to characterize their performance. Ultimately, we aim to create a *Tribo Tribe* material library as a reference for other designers and researchers. We plan to engage with industrial and interaction designers to explore how they might apply these materials for tangible interactions. This will help us surface design guidelines for *Tribo Tribe*.

Furthermore, we aim to address *Tribo Tribe*'s limitations: 1) a shielding case with aluminium foils or dehumidifiers will be designed to absorb environmental noise and humidity; 2) before the signal test, target objects will be moved several times to accumulate the electrons of material pairs as a warm-up session; 3) contact-less movements are beneficial to reducing material abrasion. Therefore, an in-depth technical evaluation will be undertaken to uncover factors affecting signal strength.

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